

Cyber Threat Intelligence (CTI)

Course Notes

Foundation Threat Intelligence Analyst

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1. Cyber Threat Intelligence: meaning and purpose

CTI is threat-focused cyber security. It helps organisations understand adversaries and make better defensive decisions.

Cyber Threat Intelligence (CTI) is the collection, processing and analysis of information about current or potential cyber threats to an organisation. The aim is to produce insight that helps decision-makers understand risk and take action before, during or after an incident.

- **Threat-centric:** CTI focuses on the threat, the attacker and how they may harm the organisation.
- **Action-oriented:** Good CTI should support a decision or action, such as blocking an indicator, prioritising a patch, changing policy or preparing incident response.
- **People-focused:** CTI is not only about technical data. It is also about understanding people, motivations, behaviours and decision-making.
- **Emerging field:** CTI is an important and developing specialism within wider cyber security.

Key definition

Cyber Threat Intelligence is insight that an organisation uses to understand the threats it faces and to reduce the harm an adversary may cause.

2. Core concepts: cyber, threat and intelligence

Term	Note
Cyber	A widespread, interconnected digital environment. It includes more than the public internet, such as intranets, Internet of Things (IoT), surface web, deep web and dark web.
Threat	A person, group, event or thing with the ability and potential to cause harm or create negative consequences for a victim.
Intelligence	Processed and analysed information that supports decision-making. In cyber security, intelligence helps organisations understand threats and improve their security posture.

Threat in simple terms

- A threat can be a statement of intent to cause harm, a warning sign that harm may occur, or a person or thing that may make a situation unsafe.
- Not all threats involve violence. In cyber security, a threat may involve theft, disruption, extortion, espionage, data exposure or reputational damage.
- A security threat can come from people, technology, weak processes, business logic flaws or external adversaries.

3. Data, information, intelligence and IOCs

A CTI analyst turns raw data into useful intelligence through collection, context and analysis.

Term	Note
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Data	Simple facts that are often available in large quantities. In cyber security this could include IP addresses, domain names, file hashes, timestamps or system logs.
Information	Data that has been collected, organised and given useful context.
Intelligence	Information that has been processed, analysed and interpreted so it can inform decision-making.
Indicator of Compromise (IOC)	A piece of forensic evidence that may indicate malicious activity on a system or network. Examples include suspicious IP addresses, domains, file hashes, registry keys or log entries.
IOC feed	A stream or list of IOCs that can feed security controls, monitoring systems or analysts. An IOC feed alone is not full intelligence unless it is enriched with context and actionability.

Simple intelligence flow

1. Collect raw data from relevant sources, such as logs, OSINT, internal tools or threat feeds.
2. Organise and validate the data to create information.
3. Analyse the information to identify patterns, relevance, impact and meaning.
4. Produce intelligence that supports a decision or practical security action.

4. Why CTI is useful to organisations

CTI helps organisations move from reactive security to more proactive risk management.

- Identify and assess potential threats to networks, systems, users and business assets.
- Improve the security posture by taking proactive measures to reduce the likelihood or impact of attacks.
- Support incident response by providing current knowledge about known threats, attacker behaviour and likely objectives.
- Prioritise patching and vulnerability management based on risk, likelihood and potential impact.
- Monitor external sources for signs of compromise, planned attacks or exposure of organisational data.
- Understand tactics, techniques and procedures (TTPs) used by malicious actors.
- Explain why an attacker may target the organisation, which can highlight vulnerable systems, processes or data.

CTI across the business

Term	Note
Security Operations Centre (SOC)	Uses CTI and IOCs to support monitoring, detection, blocking and security controls such as firewalls and SIEM rules.
Incident response	Uses CTI to understand what happened, assess attacker behaviour and support attribution where

	appropriate.
Patch and vulnerability management	Uses threat intelligence to prioritise fixes for vulnerabilities that are being exploited or are most relevant to the organisation.
Human resources and policy	Uses CTI to support awareness, crisis planning and internal policy decisions.
Business risk management	Uses CTI to understand how cyber threats may affect operations, reputation, finance and legal exposure.
Strategic level	Supports the board and senior decision-makers with high-level risk and threat understanding.
Operational level	Supports SOC operations, control tuning and security process improvements.
Tactical level	Supports threat hunting, log analysis, detection engineering and technical investigations.

5. The role of the CTI analyst

The CTI analyst combines analytical judgement, technical understanding and communication skills to create useful insight.

A CTI analyst gathers data, evaluates threats and reports on risks that could affect an organisation. The analyst combines knowledge of the wider threat landscape with analytical skills to support decisions and reduce risk.

- Identify organisational intelligence requirements.
- Collect relevant data and conduct all-source analysis.
- Identify, monitor and assess potential threats or weaknesses.
- Validate whether security requirements and assumptions are being met.
- Create reports that explain key findings to security teams and other stakeholders.
- Present findings clearly and propose countermeasures to reduce risk.
- Produce both short-term assessments and longer-term evaluations.

Art, craft and science of CTI

- **Art:** Judgement, curiosity, critical thinking and understanding human behaviour.
- **Craft:** Practical tradecraft, communication, structured analysis and report writing.
- **Science:** Technical evidence, repeatable methods, data analysis and validation.

6. Hackers, threat actors and motivations

The word hacker is not always negative. Context, consent and intent matter.

Term	Note
Hacker	A person who uses technical skills to overcome a problem. The word can be positive or negative depending on consent and intent.

Black hat	Conducts hacking activity without the owner's consent and with malicious intent.
White hat	Conducts hacking activity with permission from the technology owner, usually to improve security.
Grey hat	Acts without permission but usually stops short of clear malicious activity. This is still risky and may be unlawful.
Script kiddie	A low-skilled attacker who uses existing tools or scripts without deep understanding.
Advanced Persistent Threat (APT)	A highly capable and often long-term threat actor or campaign that is persistent, targeted and resourceful.

Examples mentioned in the notes

- Tim Berners-Lee - associated here with the white-hat or positive technology camp because of his contribution to the World Wide Web.
- Gary McKinnon - gained access to military networks and claimed he was searching for information about a UFO cover-up.
- Michael Calce, also known as MafiaBoy - associated with high-profile denial-of-service attacks.
- Phineas Fisher - noted as a hacker example for further reading.
- Stuxnet - noted as a significant cyber case study for CTI learning.
- Digital Pearl Harbour - a phrase used to describe fears of a major cyber event with serious national impact.

The unholy trinity of threat actors

Term	Note
Nation-state backed actors	Often seek intelligence, strategic advantage, disruption or intellectual property that supports political or national objectives. They may operate with protection from their home state.
Cyber criminals	Usually seek financial gain by stealing money, data or access that can be sold or monetised.
Hactivists	Use hacking to support social or political objectives, create disruption or draw attention to a cause.

The line between cyber criminals and nation-state activity can become blurred. Criminal groups may be tolerated, supported or directed by states, and nation-state actors may use criminal methods or infrastructure.

7. The CIA triad

The CIA triad is a core model for thinking about secure systems.

Term	Note
Confidentiality	Keeping data private and ensuring only authorised people, systems or processes can access or modify it.
Integrity	Ensuring data is accurate, reliable and protected from unauthorised or accidental alteration.
Availability	Ensuring systems, networks, applications and data are available when needed.

Different organisations may prioritise these principles differently. For example, a healthcare system may place heavy emphasis on availability during emergencies, while a legal or intelligence organisation may focus strongly on confidentiality.

8. Key security and CTI terminology

Term	Note
Threat vector	The path or method a threat uses to gain access to a victim's computer, network, system or data.
Vulnerability	A flaw in people, technology, process or business logic that a threat can exploit to create an effect on the victim.
Security control	A countermeasure designed to detect, prevent, reduce or counteract security risk.
Attack surface	The total set of points an attacker could use to attempt compromise or disruption.
Watering hole attack	An attack where adversaries compromise a website or service that the intended victims are likely to visit.
VPN	Virtual Private Network. Often used to provide remote access or a secure connection into a network.
RDP	Remote Desktop Protocol. A remote access technology that can become a major risk if exposed or poorly secured.
Software vulnerability	A weakness or flaw in software that can be exploited.
CVE	Common Vulnerabilities and Exposures. A standardised way to identify and record publicly known vulnerabilities.
DDoS attack	Distributed Denial-of-Service attack. An attempt to overwhelm a service, system or network so legitimate users cannot access it.

TTP	Tactics, Techniques and Procedures. The behaviours, methods and processes used by threat actors.
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9. The intelligence cycle

The intelligence cycle provides a structured way to move from a question to useful intelligence and then back to new requirements.

Term	Note
Direction	The intelligence team receives direction from the customer or stakeholder. This defines the intelligence requirements and what decisions the intelligence must support.
Collection	The team collects relevant data and information from sources and agencies. This may include internal telemetry, OSINT, threat feeds, reports or other sources.
Analysis	Information is assessed, compared, validated and interpreted to produce intelligence.
Dissemination	The finished intelligence is shared with the customer or stakeholder in a useful form. Feedback then informs new direction.

What the cycle achieves

- Defines what information is needed to support decision-making and security operations.
- Supports structured collection from relevant sources.
- Converts raw data into usable information by verifying, validating and combining it.
- Identifies patterns, trends and potential threats.
- Shares the results with stakeholders who need to act on them.
- Uses feedback to improve future collection and analysis.

Sterile corridor

A sterile corridor is a deliberate separation between phases or parts of an intelligence activity. The aim is to obscure certain details of an activity from other members of the intelligence team where separation is required.

10. Legal and ethical considerations

CTI work must be lawful, proportionate and ethically justified.

- Laws are structured rules used to govern society. Breaking the law can lead to legal consequences.
- Ethics are moral principles that guide behaviour. Organisations may also create internal ethical policies and guidelines.

- Ignoring organisational ethics may not always be illegal, but it can lead to disciplinary action, loss of trust or loss of employment.
- CTI analysts should consider legality, necessity, proportionality, authorisation and privacy when collecting or using information.

Laws mentioned in the notes

- Computer Misuse Act 1990
- Data Protection Act 1998
- Police and Justice Act 2006
- Bribery Act 2010
- Regulation of Investigatory Powers Act 2000
- Proceeds of Crime Act 2002
- Official Secrets Act 1989
- Telecommunications (Lawful Business Practice) (Interception of Communications) Regulations 2000
- Human Rights Act 1998

11. Measuring threat actors

Term	Note
Motivation	Why the actor acts. Examples include money, politics, ideology, espionage, status, revenge or disruption.
Capability	What the actor can realistically do, based on skill, resources, tools, access and persistence.
Intent	What the actor aims or plans to do against a target.

These three factors help analysts assess how likely a threat is, how serious it may be, and what defences or mitigations should be prioritised.

12. Books mentioned during the course

- Burning Chrome - William Gibson
- Rise of the Machines - Thomas Rid
- Psychology of Intelligence Analysis - Richards Heuer
- The Art of Deception: Controlling the Human Element of Security - Kevin D. Mitnick and William L. Simon, with foreword by Steve Wozniak

13. Quick revision checklist

- Can I explain CTI in one sentence?
- Can I distinguish between data, information and intelligence?
- Can I define an IOC and explain why an IOC feed alone is not the same as intelligence?
- Can I explain why CTI is useful to SOC, incident response, vulnerability management and business risk?
- Can I describe the role and responsibilities of a CTI analyst?
- Can I explain the difference between black-hat, white-hat and grey-hat hacking?

- Can I compare nation-state actors, cyber criminals and hacktivists?
- Can I explain the CIA triad with examples?
- Can I define threat vector, vulnerability, security control, attack surface, CVE, APT, TTP and DDoS?
- Can I explain the intelligence cycle: direction, collection, analysis and dissemination?
- Can I describe what makes CTI actionable?
- Can I identify legal and ethical issues that may affect CTI work?

14. Glossary of key terms

Term	Note
Actionable intelligence	Intelligence that is at the right level, enriched with context and clear enough for a decision-maker to take action.
Adversary	An attacker or hostile actor that may attempt to harm an organisation.
All-source analysis	Analysis that uses several types of sources to build a fuller picture of the threat.
Attribution	The assessment of who may be responsible for an attack or activity. Attribution should be handled carefully and based on evidence.
Capability	What a threat actor can realistically do, based on skills, tools, access and resources.
Intent	What a threat actor aims or plans to do.
Motivation	The reason a threat actor acts, such as money, politics, ideology, espionage or disruption.
OSINT	Open Source Intelligence. Intelligence collected from publicly available sources.
Security posture	The overall condition and maturity of an organisation's security defences, processes and readiness.
Threat hunting	Proactive searching through systems, logs and telemetry to find signs of malicious activity.
Threat landscape	The overall set of threats, adversaries, vulnerabilities and risks affecting an organisation or sector.